CO2xN: Porewater SO4/Cl

May 2025 Samples

2026-01-13

Contents

0.1 Run Information 2
0.2 Assess Standard Curves 3
0.3 Assess Check Standards 4
0.4 Assess Blanks 5
0.5 Assess Duplicates 7
0.6 Calculate mmol/L concentrations & salinity, add dilutions 8
0.7 Assess Analytical Spikes 9
0.8 Check if samples within the range of the standard curve 9
0.9 Check to see if samples run match metadata & merge info 10
0.10 Visualize Data by Plot 11
0.11 Export Processed Data 11

##Add Required Packages

0.1 Run Information

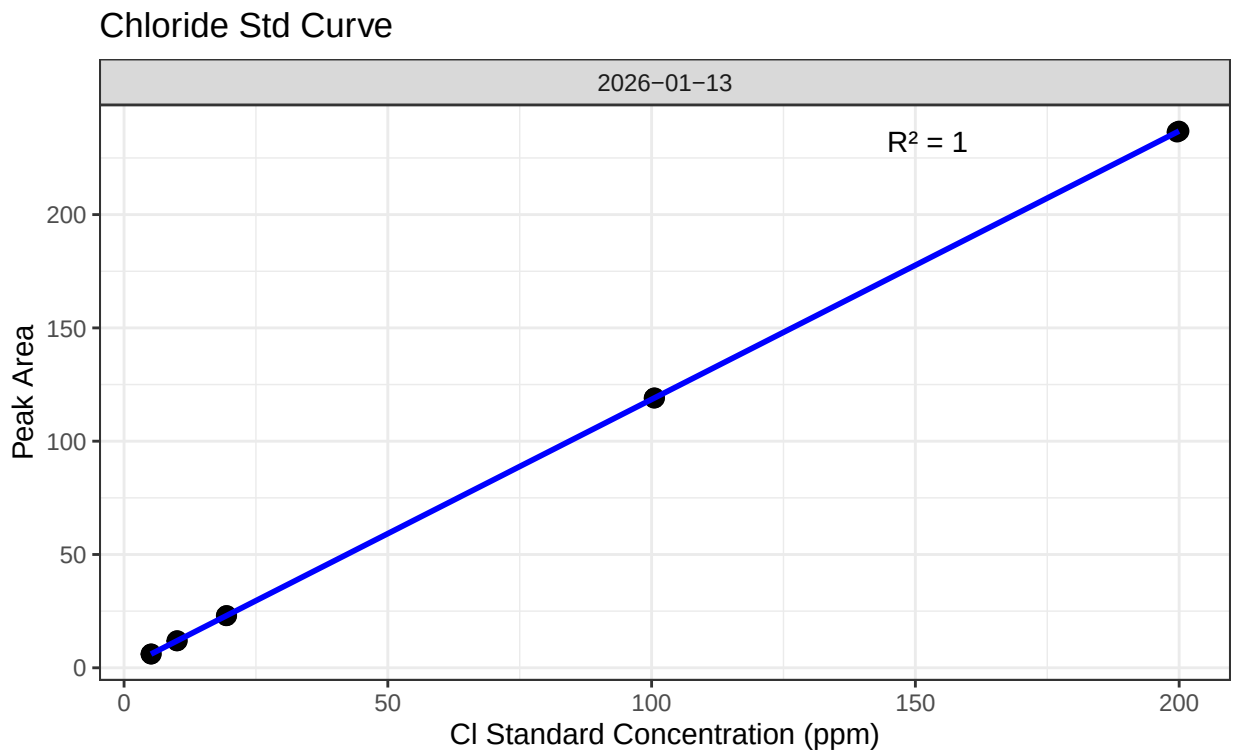
```
###### Run information - PLEASE CHANGE
  Date_Run = "2026-01-13"  #Date that instrument was run
  Run_by = "Zoe Read"  #Instrument user
  Script_run_by = "Zoe Read" #Code user
  sample_year <- "2025"
  sample_month <- "May"
  Site = "GCReW"
  Project_name = "CO2xN"
  run_notes = "
  " #any notes from the run

###### File Names - PLEASE CHANGE
#file path and name for raw summary data file
  raw_file_name_cl = "Raw Data/CO2xN_202505_Cl.txt"
  raw_file_name_so4 = "Raw Data/CO2xN_202505_SO4.txt"

#file path and name of processed data file
  processed_file_name = "Processed Data/GCReW_CO2xN_PW_Processed_Cl_SO4_202505.csv"

###### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
  Raw_Metadata = here("..", "GCReW", "GCReW_Project_Treatment_Metadata.csv")

  #qaqc log file path for this year
  Log_path = "Raw Data/COMPASS_Synoptic_Cl_SO4_QAQClog_2025.csv"
```

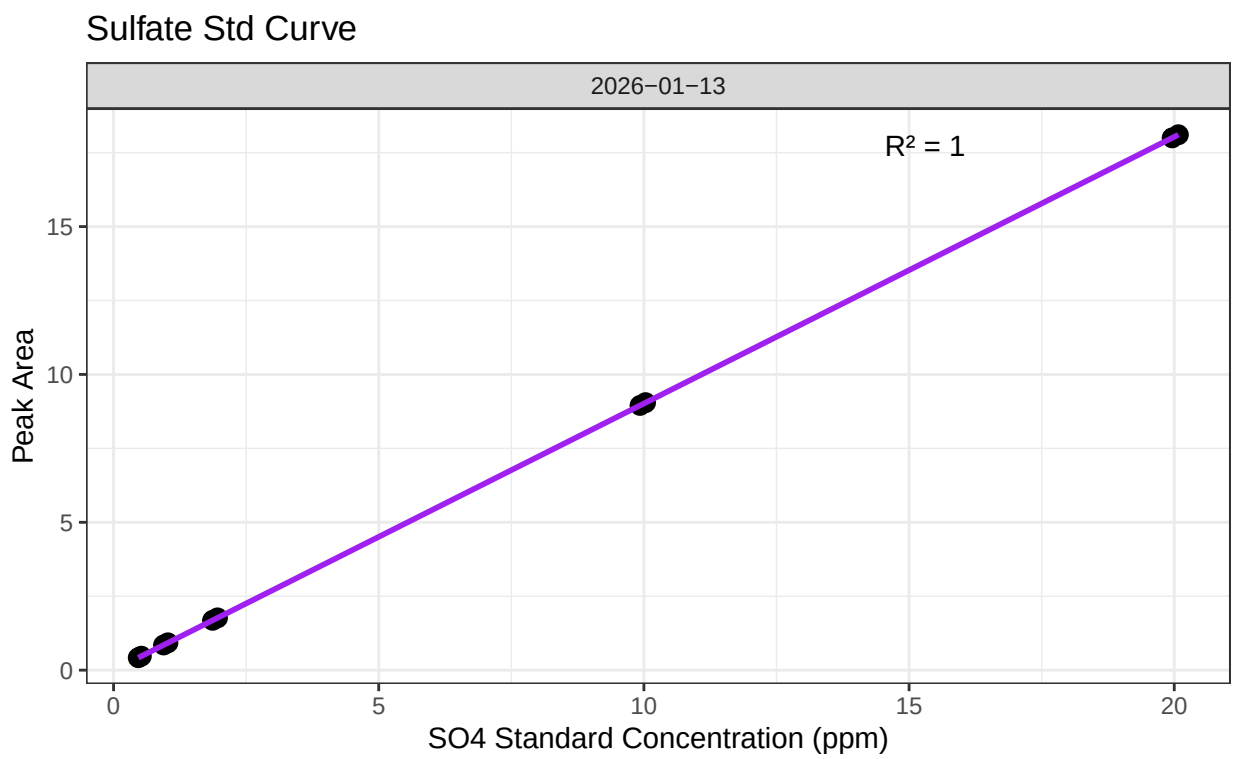
##Set Up Code - constants and QAQC cutoffs

##Read in metadata and create similar sample IDs for matching to samples

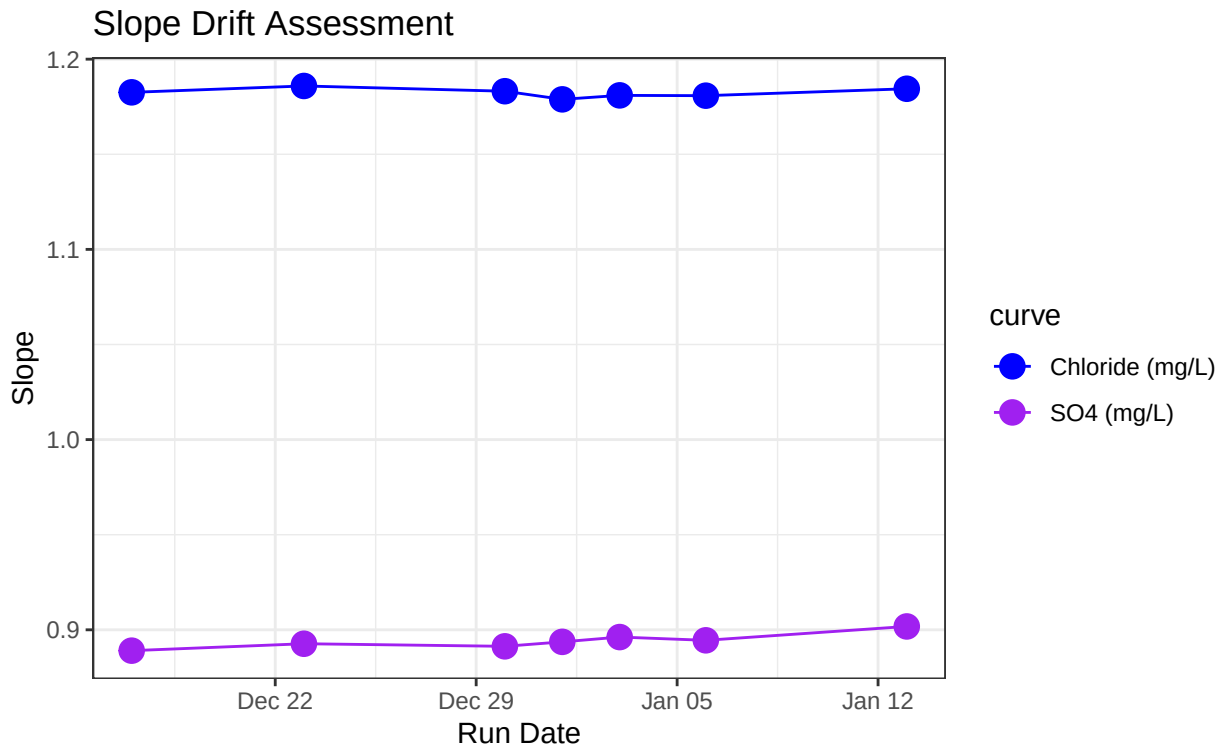
##Import Sample Data

0.2 Assess Standard Curves





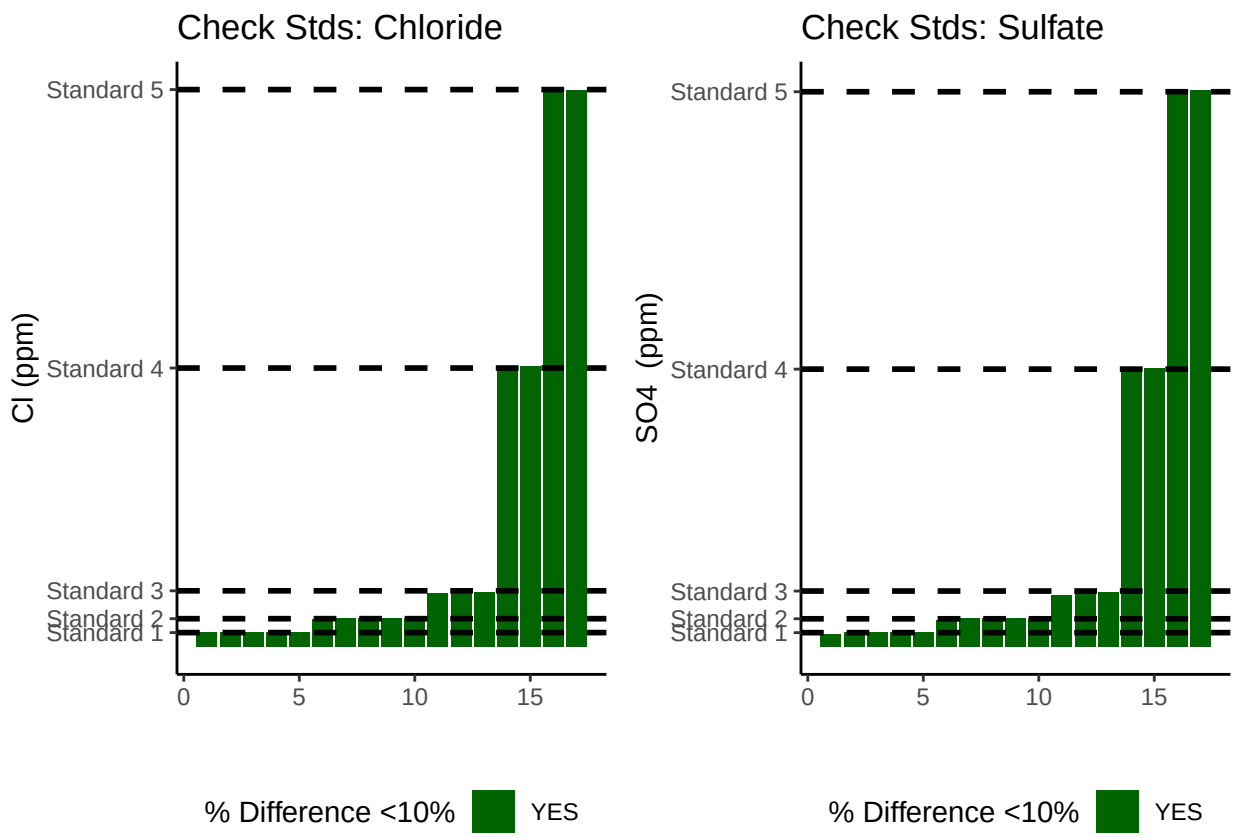

```
## [1] "QAQC log file exists and has been read into the code."
```




```
## [1] "Cl Curve r2 GOOD"

## [1] "SO4 Curve r2 GOOD"
```

0.3 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID  mean_Cl  sd_Cl    cv_Cl flag_Cl
##   <chr>        <dbl>  <dbl>    <dbl> <chr>
## 1 Standard 1    5.16 0.0300 0.00581  Chloride Check Standard RSD within Range -~
## 2 Standard 2   10.1  0.0456 0.00452  Chloride Check Standard RSD within Range -~
## 3 Standard 3   19.5  0.0707 0.00363  Chloride Check Standard RSD within Range -~
## 4 Standard 4  101.   0.0522 0.000519 Chloride Check Standard RSD within Range -~
## 5 Standard 5  200.   0.253  0.00127  Chloride Check Standard RSD within Range -~

## # A tibble: 5 x 5
##   sample_ID  mean_SO4 sd_SO4  cv_SO4 flag_SO4
##   <chr>         <dbl>  <dbl>   <dbl> <chr>
## 1 Standard 1    0.511 0.0254 0.0498  Sulfate Check Standard RSD within Range - ~
## 2 Standard 2    1.01  0.0397 0.0392  Sulfate Check Standard RSD within Range - ~
## 3 Standard 3    1.93  0.0554 0.0287  Sulfate Check Standard RSD within Range - ~
## 4 Standard 4    9.98  0.0734 0.00735 Sulfate Check Standard RSD within Range - ~
## 5 Standard 5   20.0   0.0826 0.00413 Sulfate Check Standard RSD within Range - ~

## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"

## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



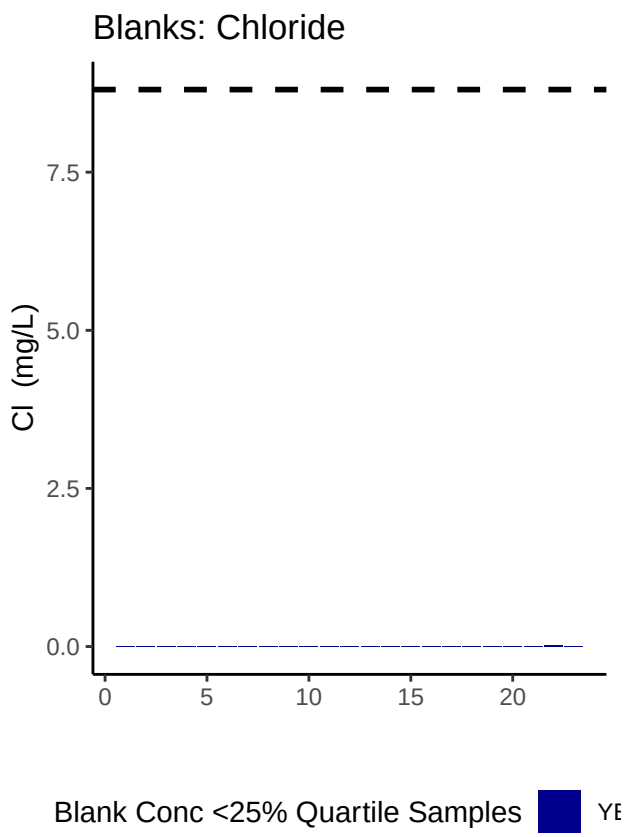

```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

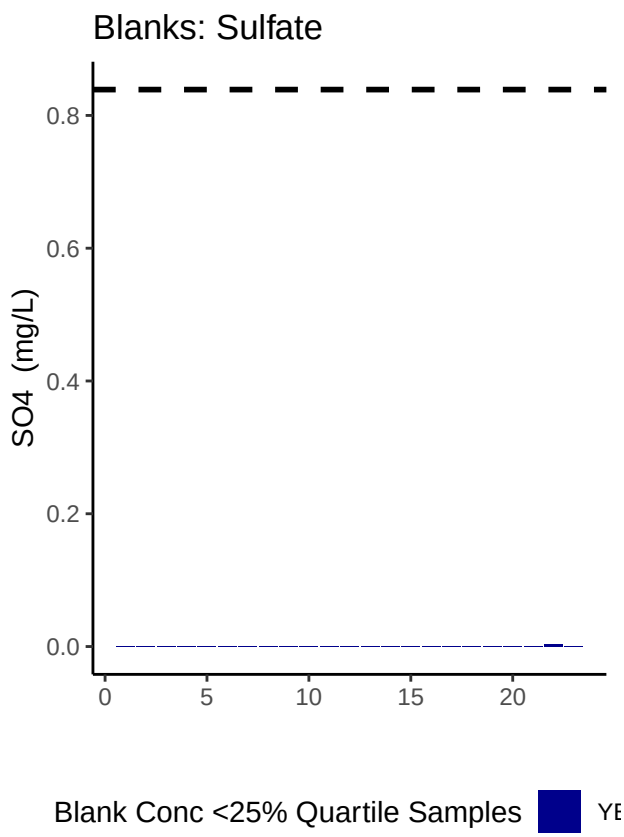
```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

0.4 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

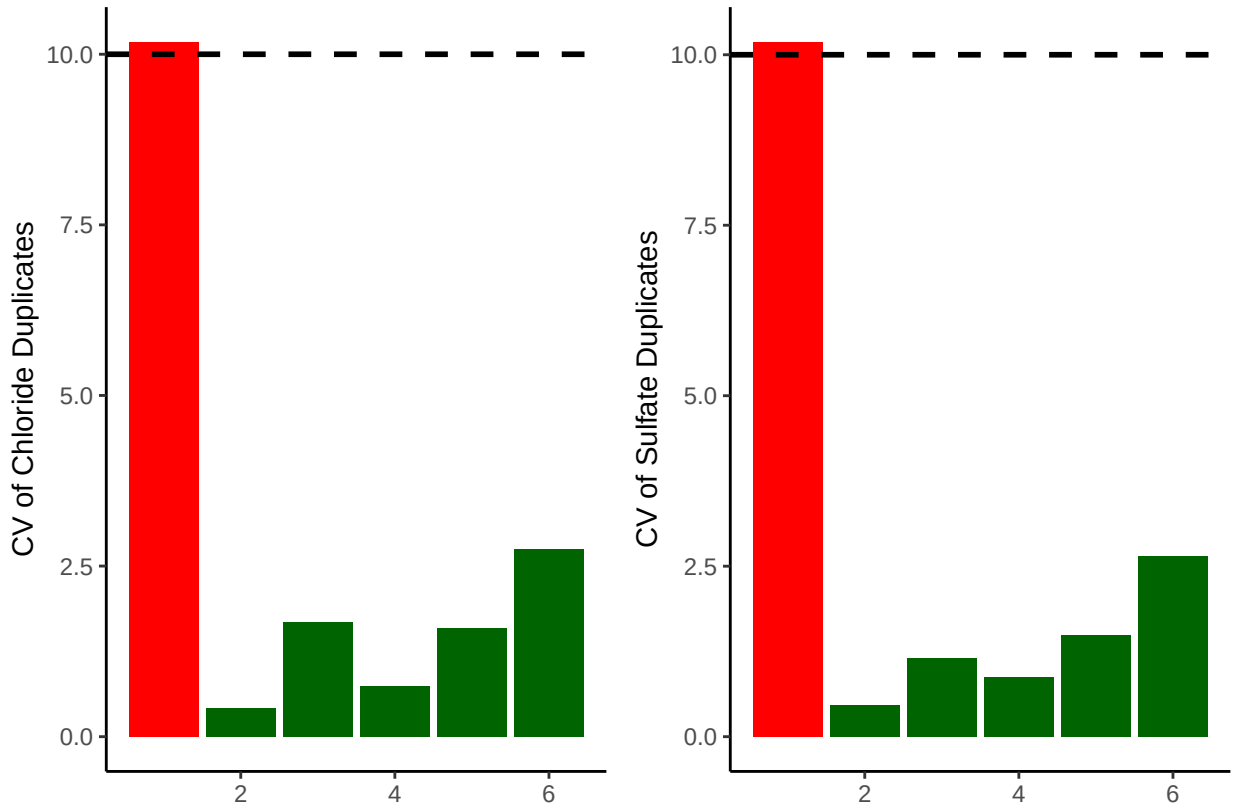
```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



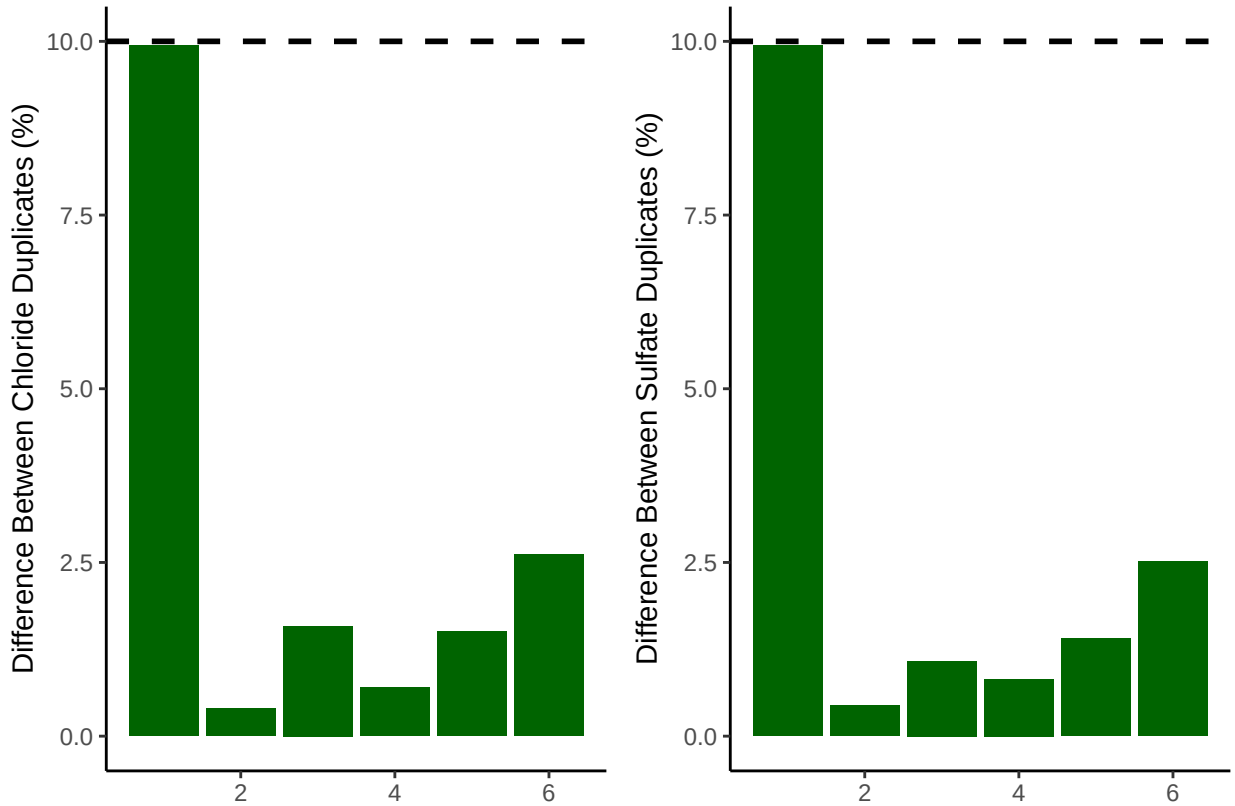



```
## Chloride blanks mean ppm:

## [1] 0.002169565

## Sulfate blanks mean ppm:

## [1] 0.0001434783
```

0.5 Assess Duplicates




```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"

## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```



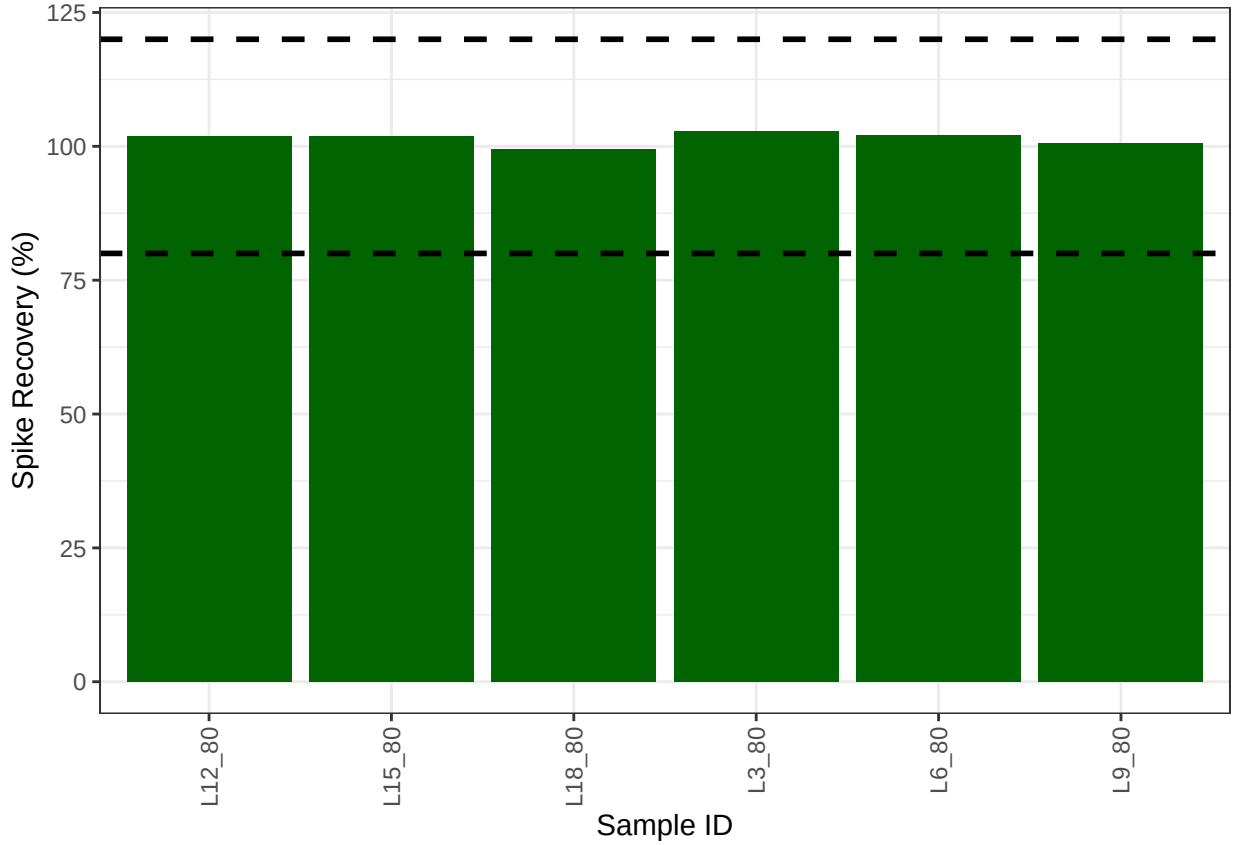

```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

0.6 Calculate mmol/L concentrations & salinity, add dilutions

```
# Convert ppm to mmol/L
all_dat$SO4_Conc_mM <- (all_dat$SO4_ppm / s_mw)
all_dat$Cl_Conc_mM <- (all_dat$Cl_ppm / cl_mw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# =((1.807*Cl_ppm)+0.026)/1000
all_dat$salinity <- ((1.8070 * all_dat$Cl_ppm) + 0.026) / 1000

#Need to determine dilution factors for your samples
#for Steph / COMPASS this depends on the site so...
all_dat$Dilution <- Dilution
# head(all_dat)
```

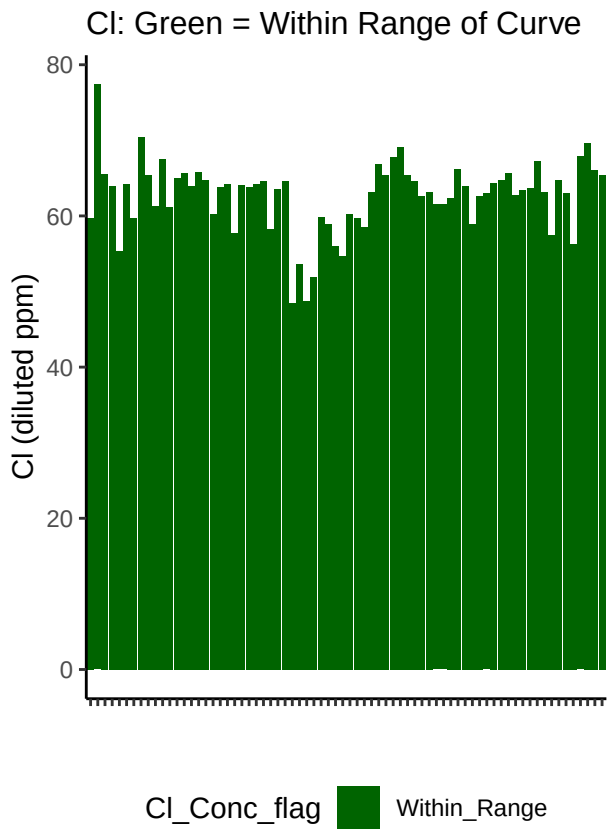
0.7 Assess Analytical Spikes




```
## [1] ">80% of SO4 spikes have a recovery between the high and low cutoff - PROCEED"
```

0.8 Check if samples within the range of the standard curve

```
## Sample Flagging
```



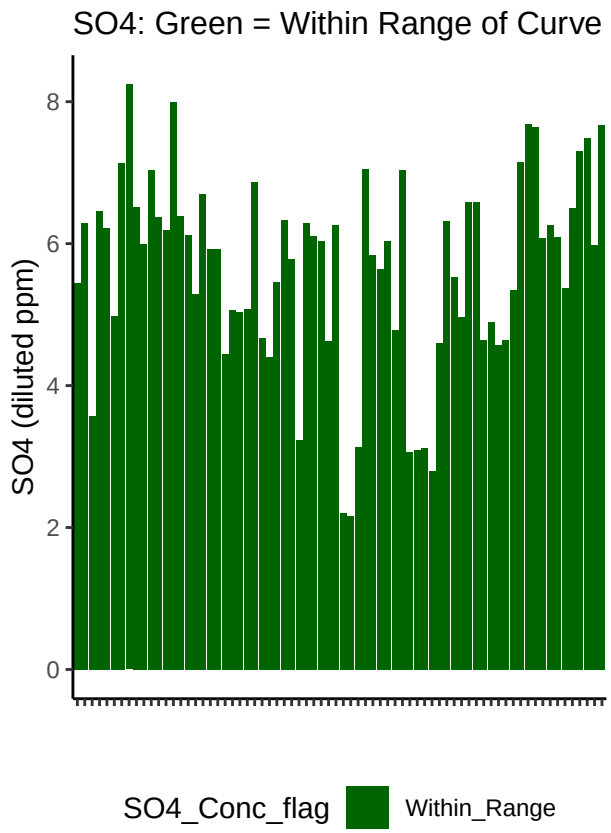


Table 1: SO4 samples

SO4_Conc_flag	Percent_samples
Within_Range	100

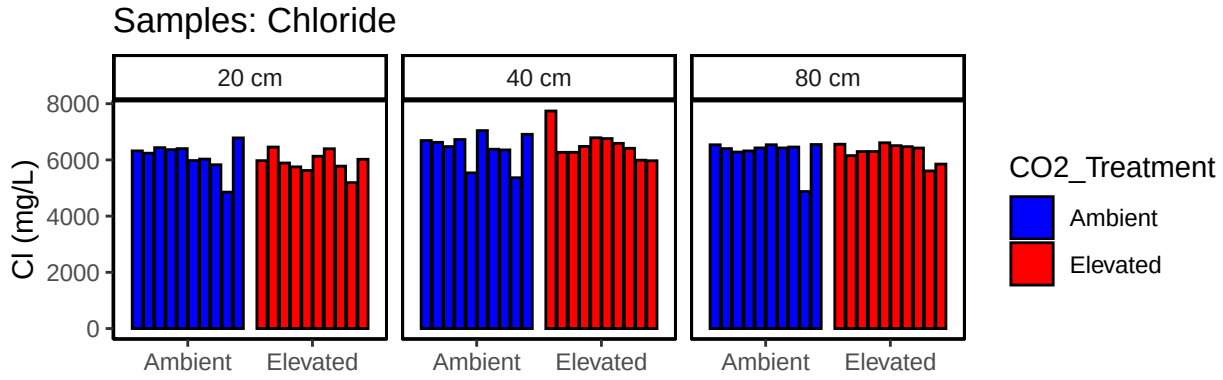
Table 2: Cl samples

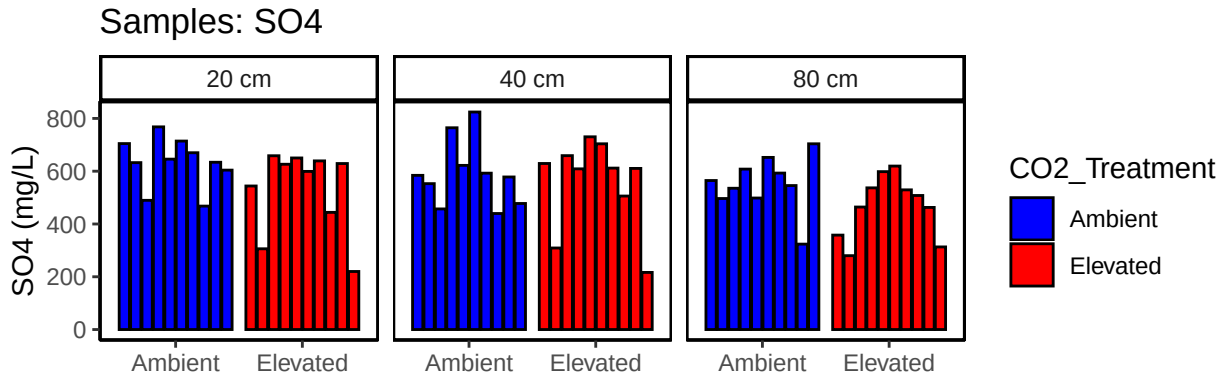
Cl_Conc_flag	Percent_samples
Within_Range	100

0.9 Check to see if samples run match metadata & merge info

```
## All sample IDs are present in metadata.
```

0.10 Visualize Data by Plot





0.11 Export Processed Data

#end